“Controlling For” other Variables.

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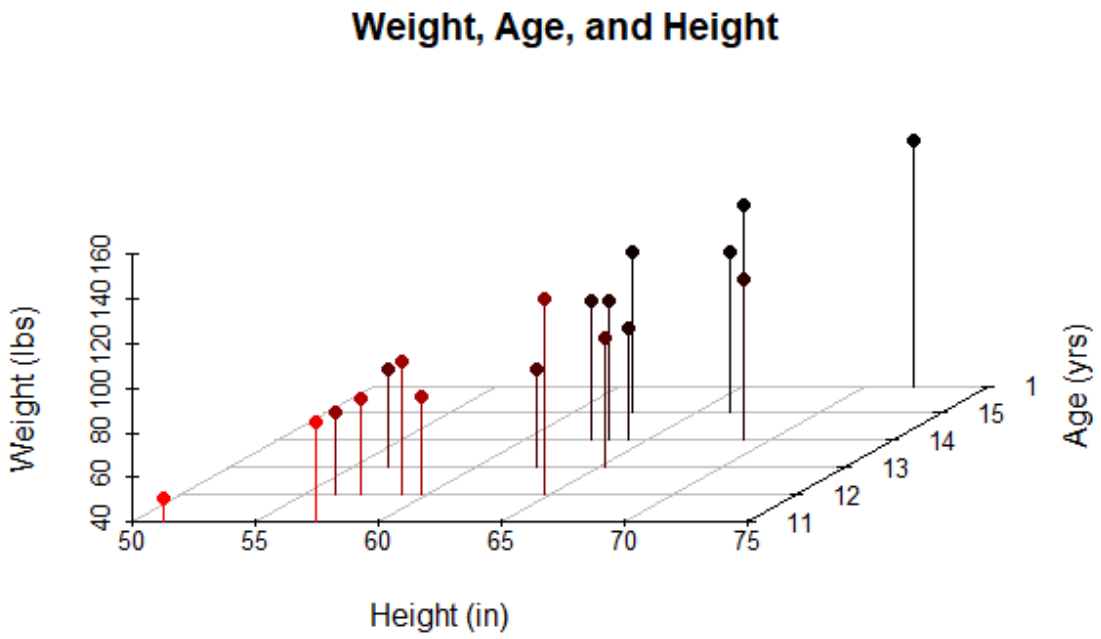
Unpacking a multi-variable relationship

- For this example, we will look at the relationship between students' heights, weights, and ages presented in the chapter.

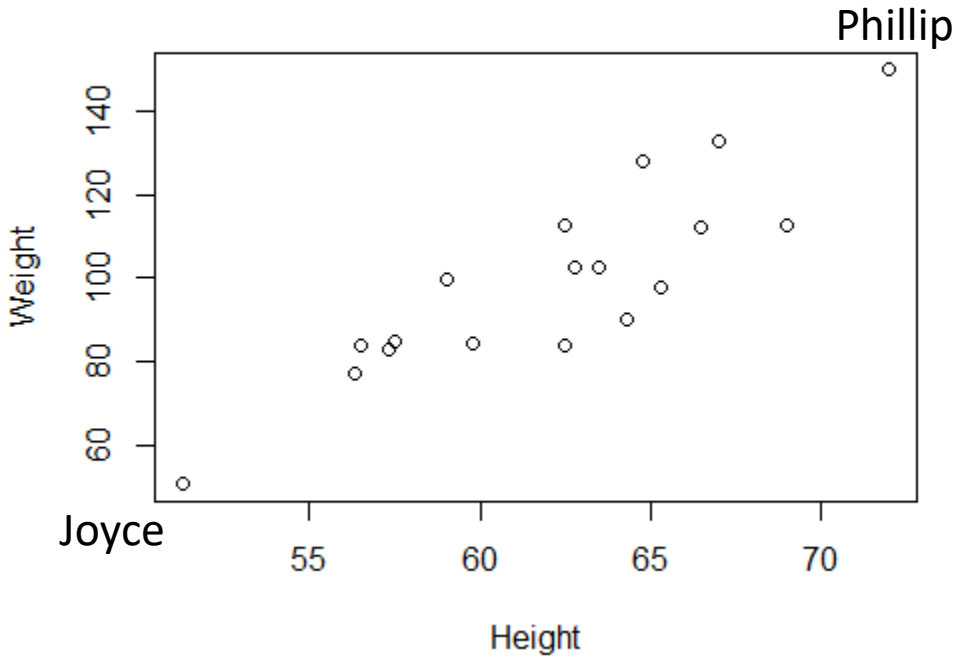
```
> head(dat)
     Name Sex Age Height Weight
1  Alfred   M  14   69.0  112.5
2   Alice   F  13   56.5   84.0
3 Barbara   F  13   65.3   98.0
4   Carol   F  14   62.8  102.5
5   Henry   M  14   63.5  102.5
6   James   M  12   57.3   83.0
```

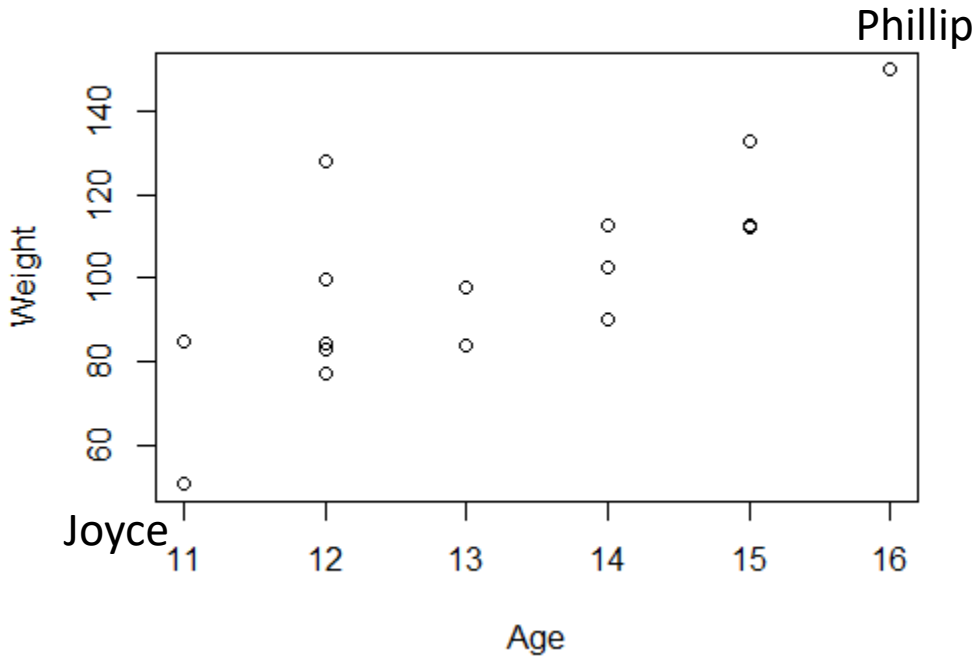


Interpreting the R output:

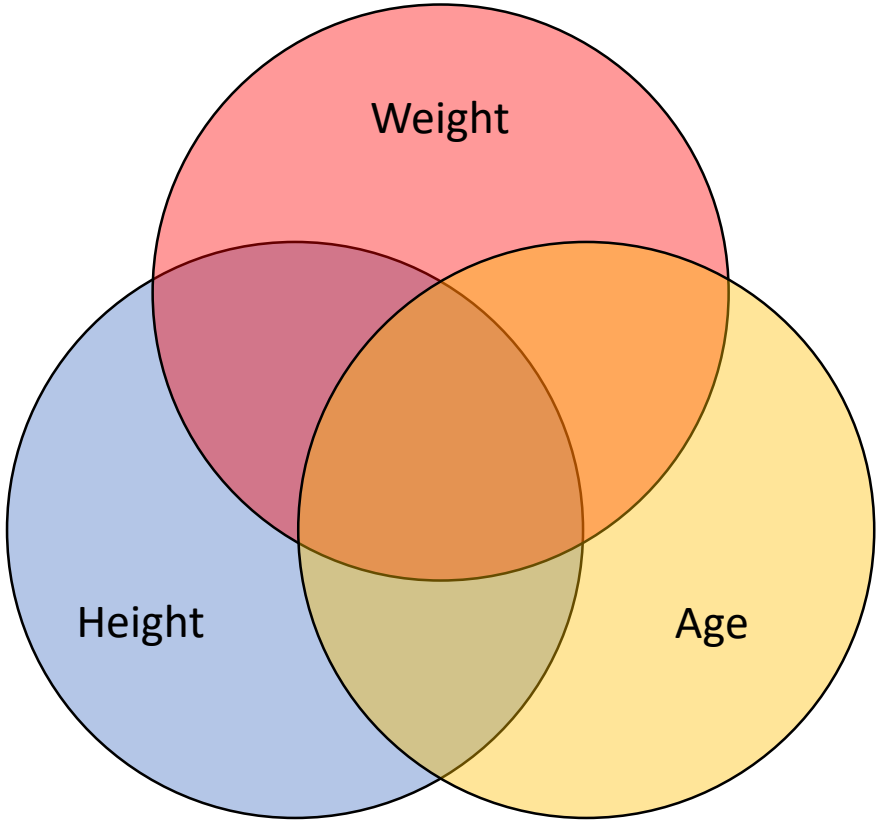
```
lm(formula = Weight ~ Height + Age, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max 
-17.962  -6.010  -0.067   7.553  20.796 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -141.2238    33.3831  -4.230 0.000637 ***
Height         3.5970     0.9055   3.973 0.001093 ** 
Age            1.2784     3.1101   0.411 0.686492    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11.51 on 16 degrees of freedom
Multiple R-squared:  0.7729, Adjusted R-squared:  0.7445 
F-statistic: 27.23 on 2 and 16 DF,  p-value: 7.074e-06
```

Plot of Weight as a Function of Height and Age

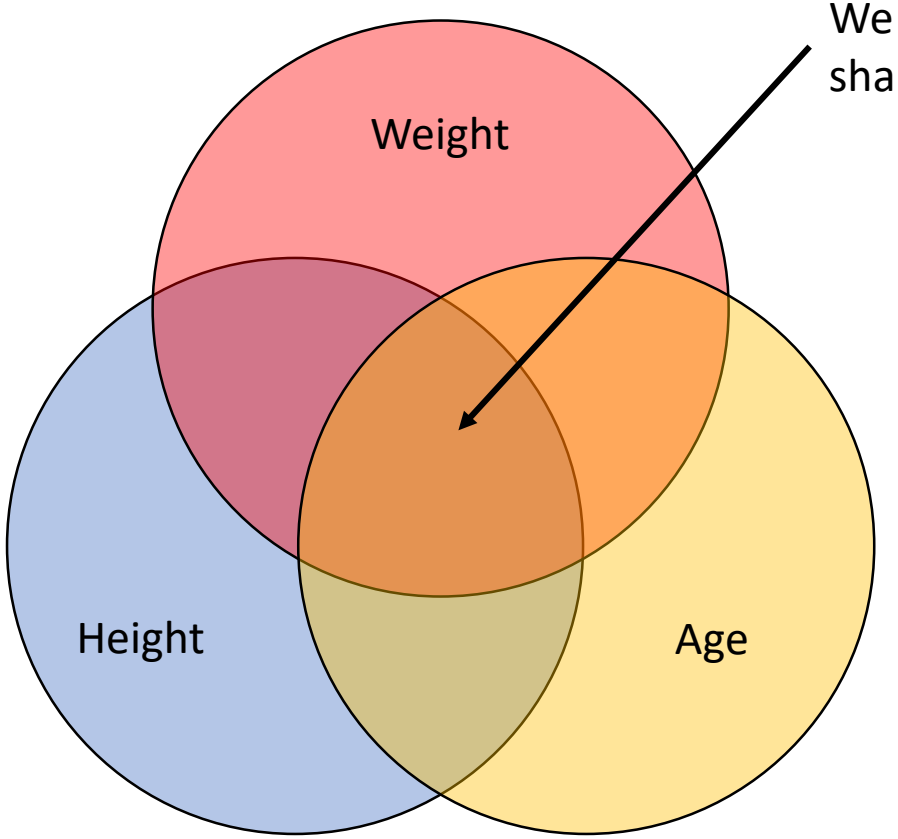




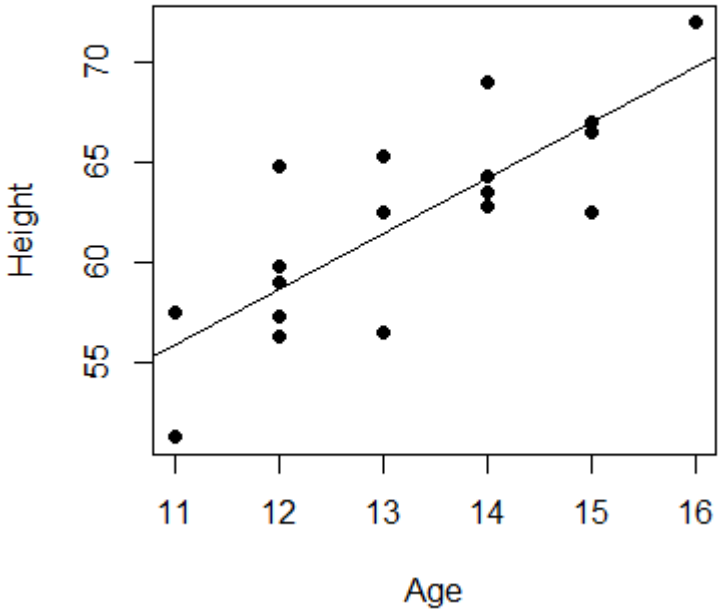
Height, Controlling for Age

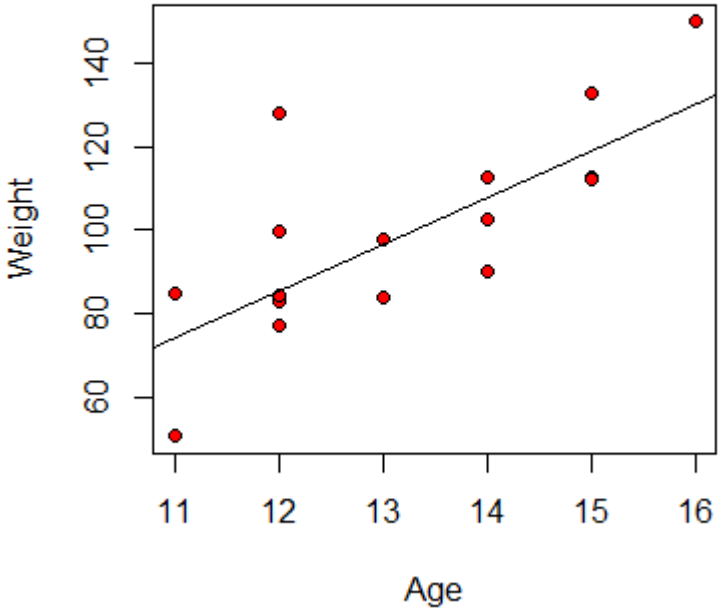


Height, Controlling for Age



We need to remove the shared variance.





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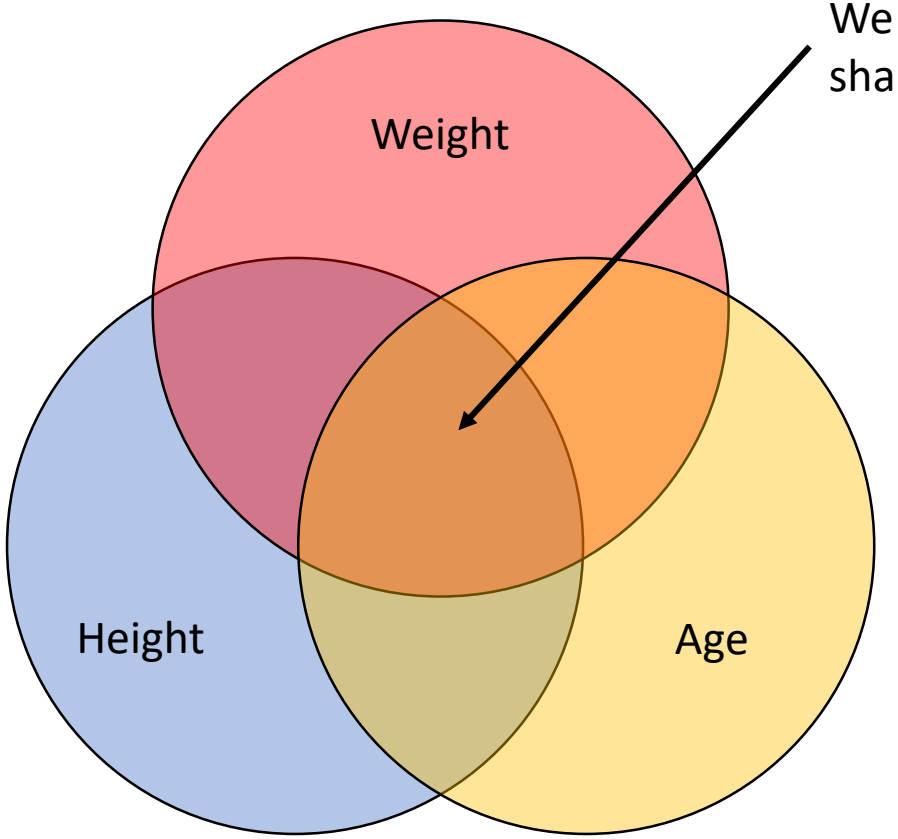
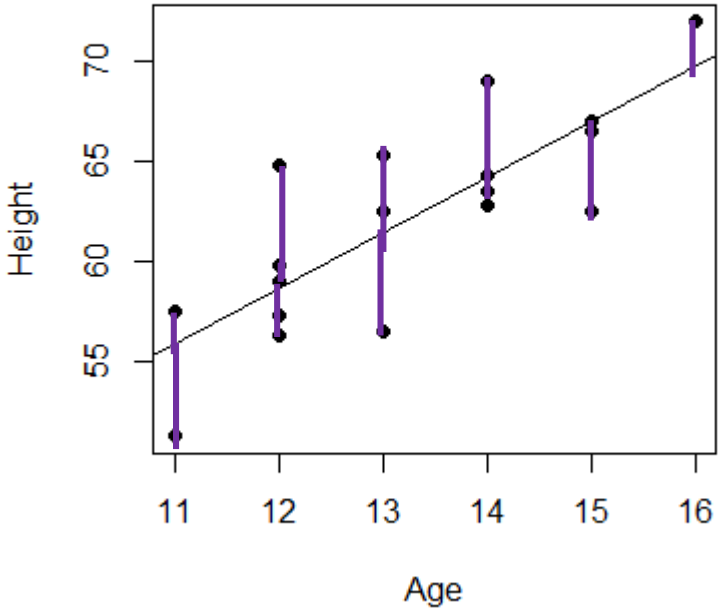
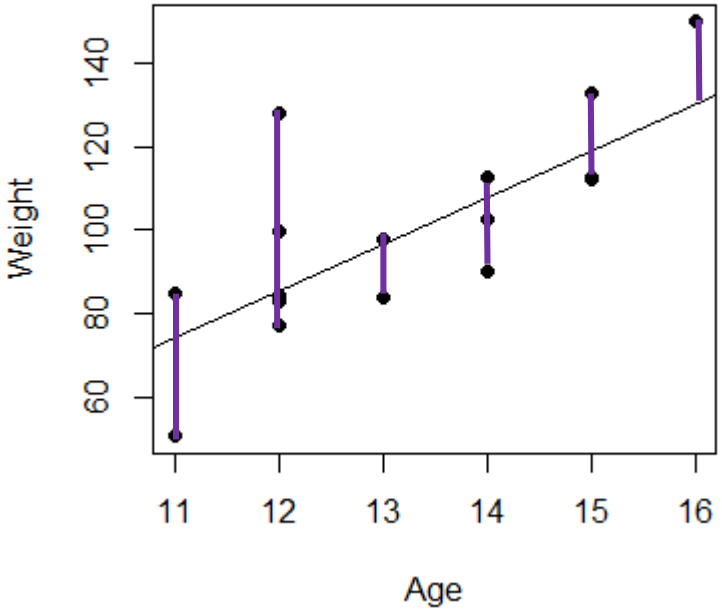


We need to remove the shared variance.

To get the parts height and weight that are independent of age, we need to extract the **residuals** from these models.

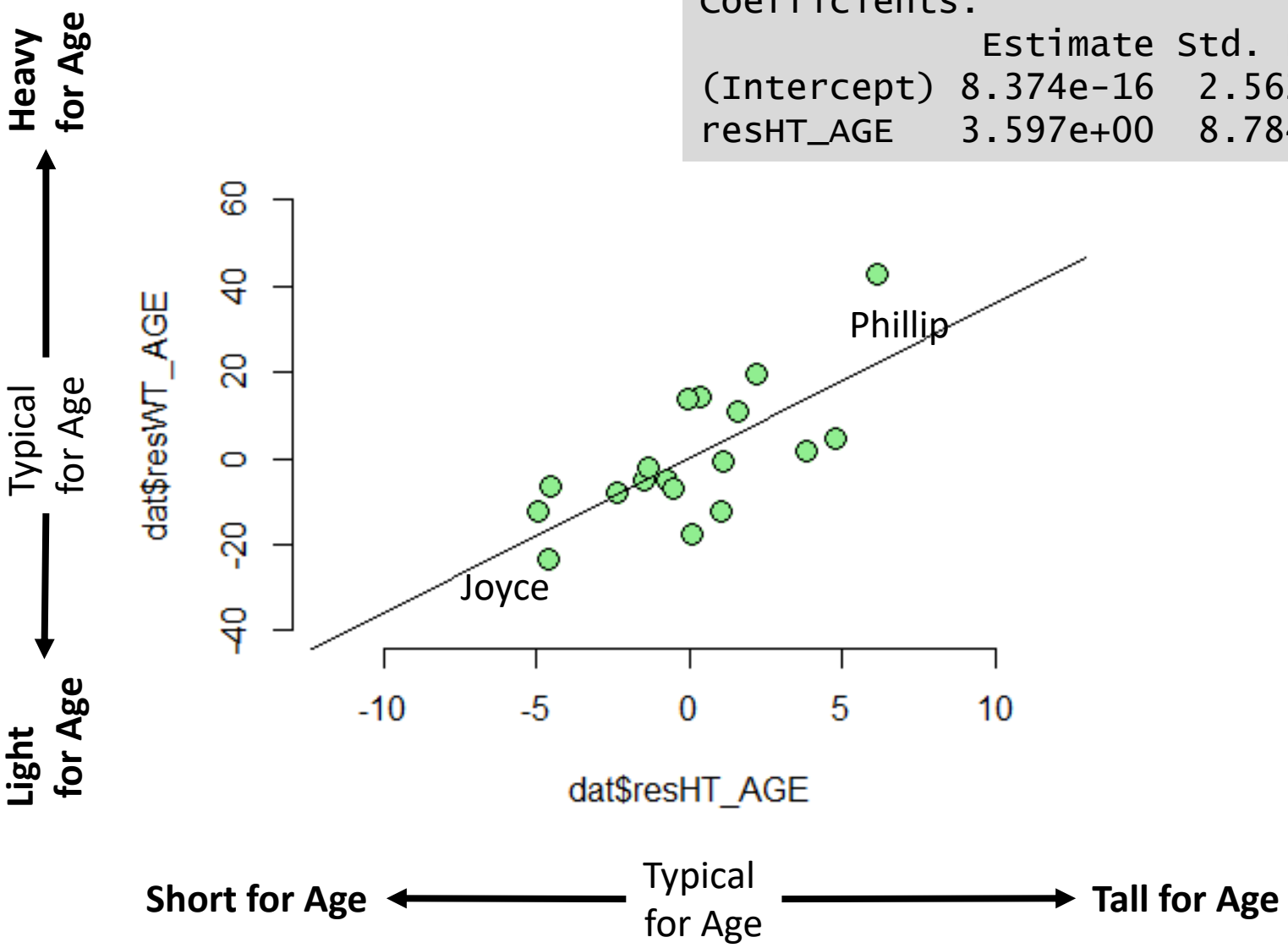
$$y_i = b_0 + b_1(x_i) + \varepsilon_i$$





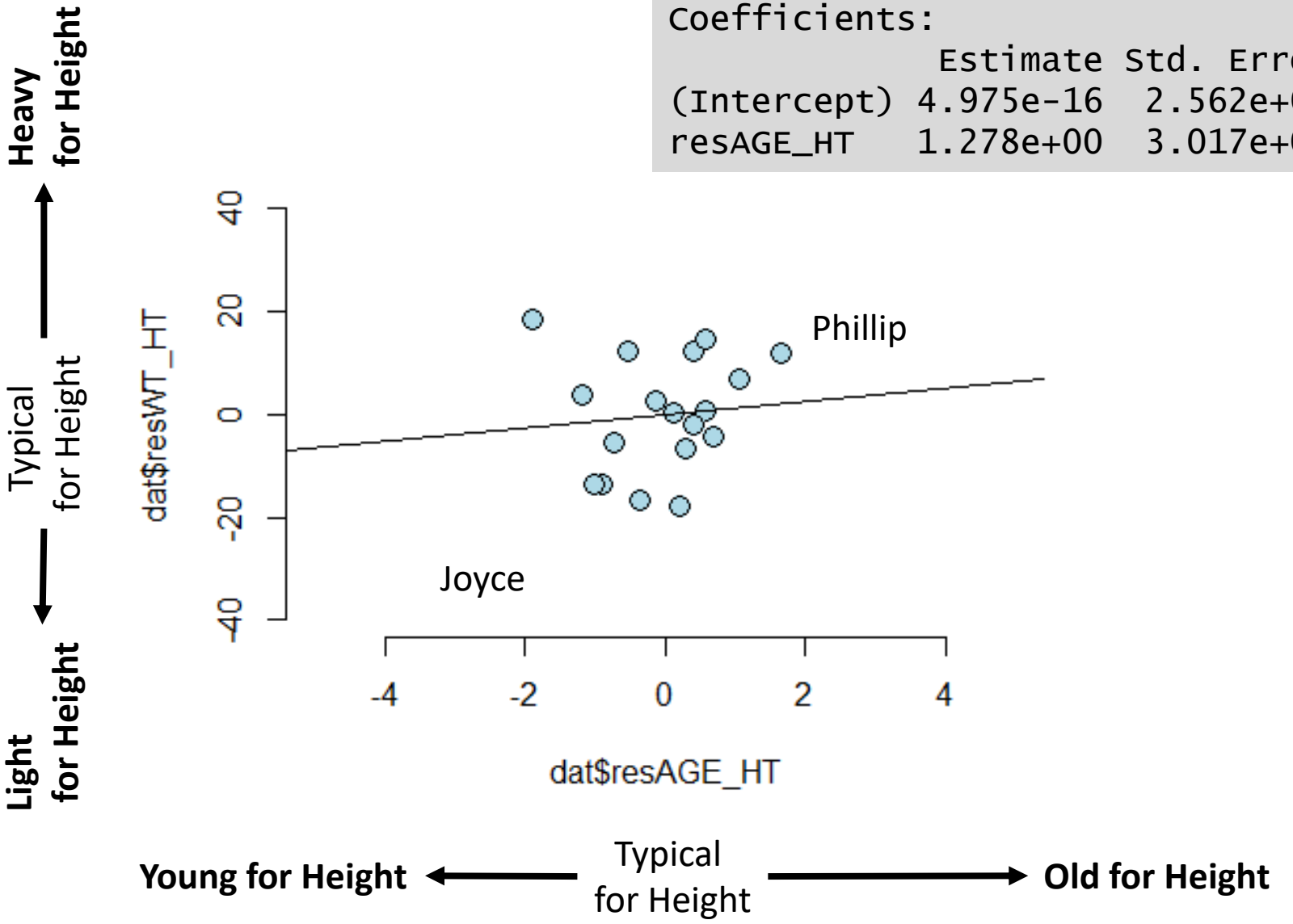
Plot of Residuals Controlling for Age

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 8.374e-16  2.562e+00   0.000 1.000000
resHT_AGE   3.597e+00  8.784e-01   4.095 0.000755 ***
```



Plot of Residuals Controlling for Age

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.975e-16  2.562e+00   0.000    1.000
resAGE_HT   1.278e+00  3.017e+00   0.424    0.677
```



Multivariable Regression

- Regressing these residuals onto each other gives us the same parameter estimates as the multivariable model.

```
# Full Multivariable Model
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(Intercept) -141.2238    33.3831  -4.230 0.000637 ***
Height         3.5970     0.9055   3.973 0.001093 **
Age            1.2784     3.1101   0.411 0.686492
---
```

```
# Weight residuals predicted by height residuals.
# Age variance is removed.
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
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```